Final Project Data Science Batch 60

Optimizing Cost, Speed, and Quality
through Predictive

Cracking the "**Efficiency Paradox**" in Talent Acquisition

Presented by
TriMedian

Meet Our Team

Project Manager/BA
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Business Analyst
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Data Scientist
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The Efficiency Paradox

Increased recruitment spending fails to accelerate hiring, as volume bottlenecks drive candidate fatigue and low acceptance rates

Industry Benchmarks for Recruitment Performance

44 Days

The global average time to hire is 44 days (HeroHunt)

58%

58% of candidates have declined offers due to poor hiring experiences (HeroHunt)

45 Days

45% of job seekers want to hear back after applying within 24 hours (CareerPlug)

$4700

The average cost per hire is about $4,700. (SHRM)

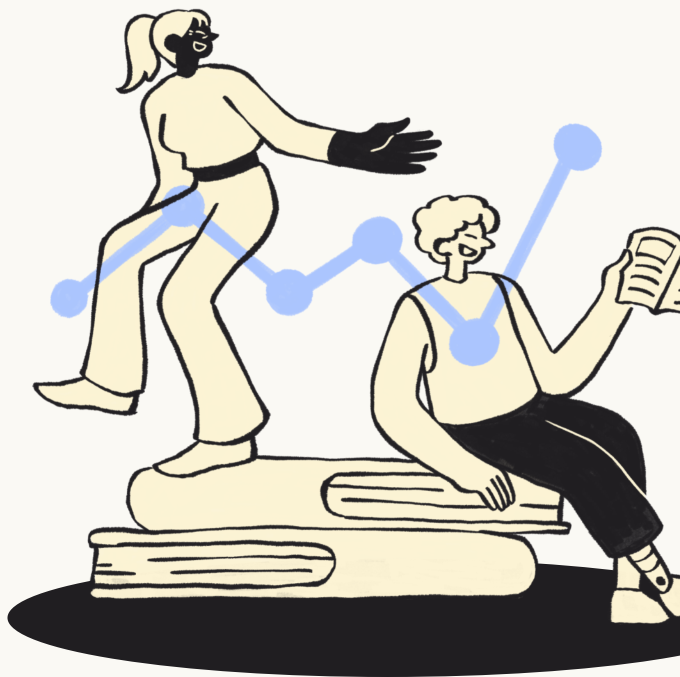
91%

91% of candidates said a positive candidate experience influenced their decision to accept an offer. (CareerPlug)

BUSINESS OBJECTIVES (The "**So What?**")

FROM INTUITION TO DATA-DRIVEN PRECISION

Increase the Offer Acceptance Rate
by identifying and predicting the key factors that influence whether candidates accept or reject job offers.



IN-SCOPE
(What We Will Deliver)

1. **Diagnostic Analytics**: Deep-dive dashboard analyzing Departmental Bottlenecks & Channel ROI.
2. **Predictive Modeling**
3. **Business Recommendations**: Actionable strategy for budget reallocation based on cost-efficiency data.

OUT-OF-SCOPE
(Constraints)

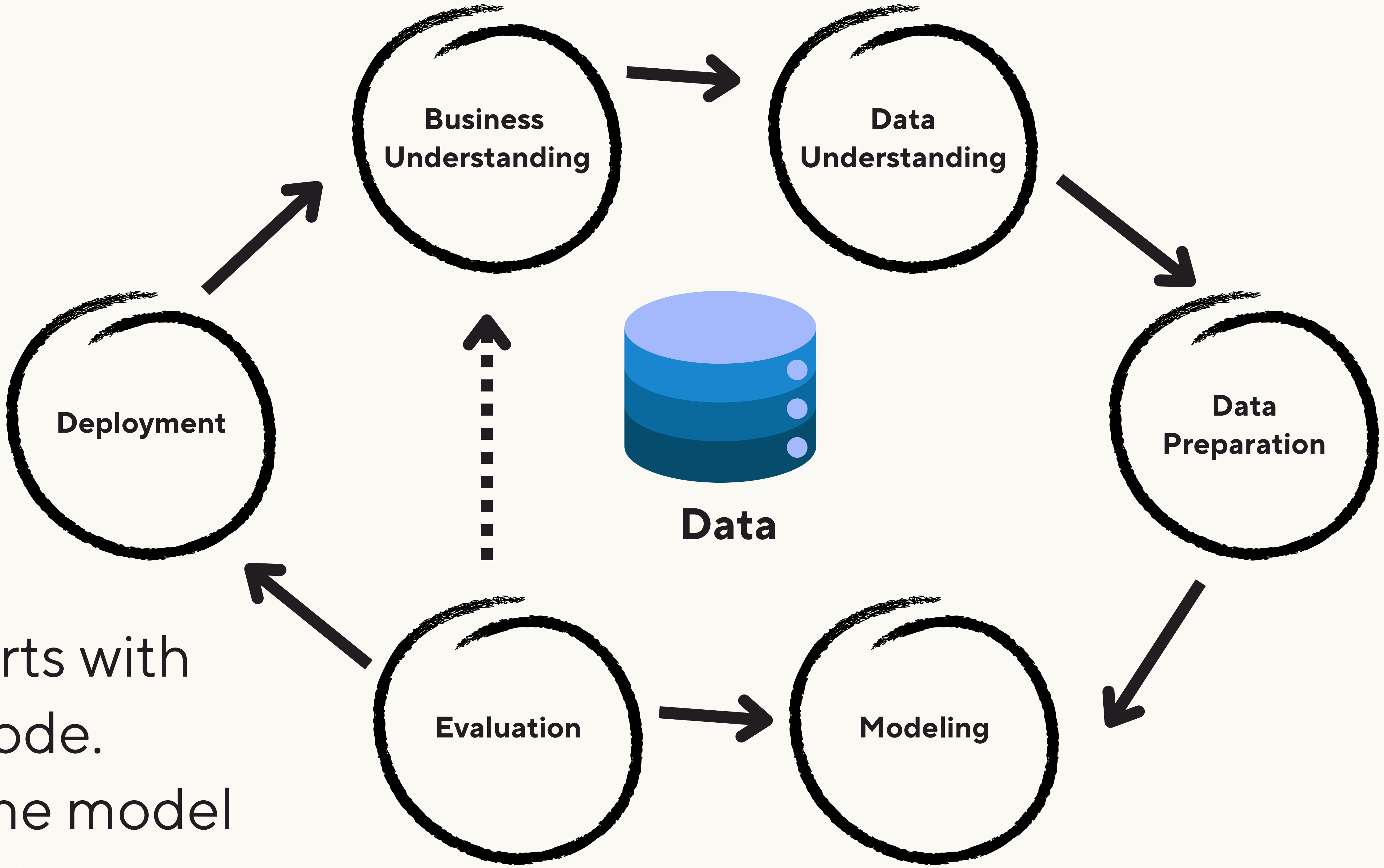
1. **Real-time Integration:** No live connection to ATS (Static CSV only).
2. **Unstructured Data**: No NLP on CVs or Video Analysis due to data limitations.

PROJECT SCOPE

STRATEGIC SCOPE & DELIVERABLES"DEFINING BOUNDARIES TO ENSURE FOCUSED EXECUTION."

METHODOLOGY (The Recipe)

EXECUTION FRAMEWORK"STANDARDIZED APPROACH USING CRISP-DM"



WHY CRISP-DM?

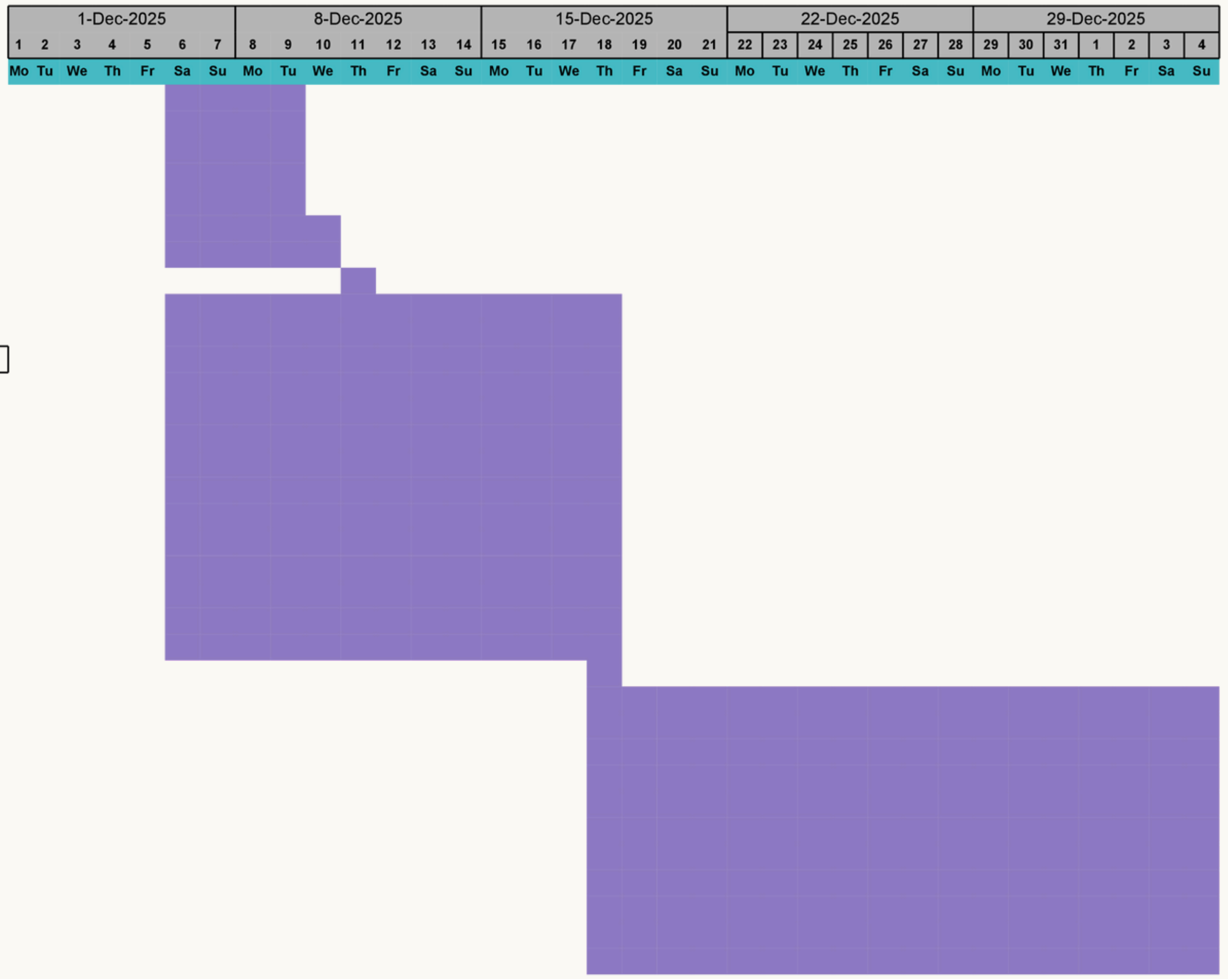
- **Business-First**: Every step starts with business objectives, not just code.
- **Iterative**: Allows us to refine the model as we understand the data better.

PROJECT ROADMAP

8-Week Sprint to Value Delivery

Start Date: 12/1/2025
Display Week: 1

Stage	PIC	ROLE	Sprint Task	Description	Plan to Start	Plan to Finish	Status	Link
0 - Project Initiation	Naufal	PM	Kickoff Review	Memahami brief, dataset, rubric, dan tujuan proyek	12/6/25	12/9/25	DONE	
	Hutomo & Naufal	BI	Industry Research	Riset talent intelligence & recruitment efficiency	12/6/25	12/9/25	DONE	
	Naufal	PM	Quick Analyst	Merumuskan problem statement, menenutkan sucess	12/6/25	12/9/25	DONE	
	Naufal	PM	Risk Assesment	Risk data, modelling, technical, timeline	12/6/25	12/9/25	DONE	
	Hutomo & Naufal	BI	Workflow	Membuat CRISP-DM diagram	12/6/25	12/9/25	DONE	
	Naufal	PM	Timeline	Membuat gantt chart & pembagian tugasa	12/6/25	12/10/25	DONE	
	Naufal	PM	Compile Proposal Deck	Membuat PPT stage 0	12/6/25	12/10/25	IN PROGRESS	
	All Team	All Team	Mentoring Session	Present stage 0 ke mentor	12/11/25	12/11/25	TO DO	Link Meet
1- Data Aquisition & Preparation	Rifki	DE	Load Raw Data	import seluruh dataset, cek struktur	12/6/25	12/18/25	TO DO	
	Rifki	DE	Data Quality Check	cek missing values, outliers, duplicates	12/6/25	12/18/25	TO DO	
	Rifki	DE	Cleaning Missing Values	Fill/remove missing values	12/6/25	12/18/25	TO DO	
	Rifki	DE	Remove Duplicates	Hapus duplikat	12/6/25	12/18/25	TO DO	
	Rifki	DE	Outlier Handling	IQR/z-score, etc	12/6/25	12/18/25	TO DO	
	Rifki	DE	Data Standarization	Normalisasi data	12/6/25	12/18/25	TO DO	
	Rifki	DE	Encoding Categorial	One-hot, label encoding	12/6/25	12/18/25	TO DO	
	Rifki	DE	Feature Engineering	Buat fitur baru	12/6/25	12/18/25	TO DO	
	Rifki	DE	Univariate EDA	Distribusi fitur	12/6/25	12/18/25	TO DO	
	Rifki	DE	Bivariate EDA	korelasi antar fitur, heatmap, dll	12/6/25	12/18/25	TO DO	
	Hutomo & Naufal	BI	Insght Sumary	BI menerjemahkan insght ke bisnis	12/6/25	12/18/25	TO DO	
	Rifki	DE	Feature Selection Strateg	Tentukan fitur untuk modeling	12/6/25	12/18/25	TO DO	
	Rifki	DE	Final Clean Dataset	Dataset siap untuk modeling	12/6/25	12/18/25	TO DO	
	Rifki	DE	EDA Report	Membuat PPT stage 1	12/6/25	12/18/25	TO DO	
	All Team	All Team	Mentoring Session	Present stage 1 ke mentor	12/18/25	12/18/25	TO DO	Link Meet
2 - Model Development	Kevin	DS	Define Modeling Strateg	Tentukan Clasification/regression, baseline metrics	12/18/25	1/10/26	TO DO	
	Kevin	DS	Train Baseline Model	Logistic baseline	12/18/25	1/10/26	TO DO	
	Kevin	DS	Train ML Model 1	melatih model a	12/18/25	1/10/26	TO DO	
	Kevin	DS	Train ML Model 2	melatih model b	12/18/25	1/10/26	TO DO	
	Kevin	DS	Train ML Model 3	melatih model c	12/18/25	1/10/26	TO DO	
	Kevin	DS	Train ML Model 4	melatih model d	12/18/25	1/10/26	TO DO	
	Kevin	DS	Train ML Model 5	melatih model e	12/18/25	1/10/26	TO DO	
	Kevin	DS	Hyperparameter TUnning		12/18/25	1/10/26	TO DO	
	Kevin	DS	Cross Validation	K-Fold Experiment, etc	12/18/25	1/10/26	TO DO	
	Kevin	DS	Model Ranking	Bandingkan semua model	12/18/25	1/10/26	TO DO	
	Kevin	DS	Model Report	Membuat PPT stage 2	12/18/25	1/10/26	TO DO	
	All Team	All Team	Mentoring Session	Present stage 2 ke mentor			TO DO	Link Meet
3 - Model Evaluation	Kevin	DS	Compute Metrics		1/10/26	1/17/26	TO DO	
	Kevin	DS	Confusion Matrix		1/10/26	1/17/26	TO DO	
	Kevin	DS	ROC	Model perfomance curves	1/10/26	1/17/26	TO DO	
	Kevin	DS	SHAP Analysis	Feature importance	1/10/26	1/17/26	TO DO	
	Hutomo & Naufal	BI	BIAS & Fairness Check	Check bias ke gender/role/etc	1/10/26	1/17/26	TO DO	
	Hutomo & Naufal	BI	BUsiness Interpretation	Translansi insght model	1/10/26	1/17/26	TO DO	
	Kevin	DS	Model Evaluation Repor	Membuat PPT stage 3	1/10/26	1/17/26	TO DO	
	All Team	All Team	Mentoring Session	Present stage 3 ke mentor	1/10/26	1/17/26	TO DO	Link Meet
4 - Deployment & Business Integration			Choose Deployment Me	Dashboard / API / Streamlit			TO DO	
			Buat API / Model Pipeline				TO DO	
			Buat dashboard	power BI?			TO DO	
			Final Presentation Deck	Membuat PPT FInal Presentation			TO DO	
			Pitching Simluation	Simulai pitching	1/24/26	1/24/26	TO DO	
			Final Day Presentation D	Present ke judge	1/31/26	1/31/26	TO DO	



COMPREHENSIVE RISK ASSESSMENT

MANAGING UNCERTAINTY (RISK REGISTER)

ID	Risk Description	Sev	Mitigation Strategy
R01	Data **(Small Dataset Size)**: Limited historical data may lead to weak generalization and unreliable recommendations.	High	Use low-complexity models (e.g., Logistic Regression/RF) & robust Cross-Validation.
R02	Data (**Data Leakage**): Risk of using "future data" as input, creating a falsely accurate but unusable model.	High	Strict feature selection based on timeline; exclude post-event variables from input.
R03	Tech (**Overfitting**): Model memorizes training data, fails on test.	High	Apply Cross-Validation (K-Fold), L1/L2 Regularization.
R04	Tech (**Interpretability**): Model is a "Black Box" for HR users.	High	Use SHAP Values/LIME to explain feature importance.
R05	Biz (**Insight Bias**): Recommendations biased to specific depts	Med	Normalize data & perform segmented analysis.
R06	Team (**Tight Deadline**) (1 Month): Compressing Stage 0-4 into 4 weeks risks incomplete delivery.	Med	Strict MVP Approach: Focus only on the core predictive pipeline; drop complex add-ons.

Project Feasibility Overview

TECHNICAL FEASIBILITY

STATUS: ROBUST & LOW RISK

WHY: WE UTILIZE A PROVEN **PYTHON** STACK AND STATIC DATA SOURCES. NO CRITICAL DEPENDENCIES ON LIVE IT INFRASTRUCTURE.

OPERATIONAL FEASIBILITY

STATUS: HIGH USABILITY

WHY: DELIVERABLES ARE DESIGNED FOR NON-TECHNICAL USERS (HR LEADERS), FOCUSING ON STRATEGIC RECOMMENDATIONS OVER COMPLEX CODE.

ECONOMIC FEASIBILITY

STATUS: COST-EFFECTIVE

WHY: ZERO SOFTWARE LICENSING COSTS (OPEN SOURCE). THE PROJECT TARGETS DIRECT OPEX SAVINGS BY OPTIMIZING CHANNEL SPEND.

BUSINESS FEASIBILITY

STATUS: HIGH STRATEGIC VALUE

WHY: SOLVES THE CRITICAL "EFFICIENCY PARADOX." DIRECTLY IMPACTS TIME-TO-HIRE AND QUALITY OF HIRE METRICS.

The Sprint to Value Starts Now

Ready to deploy the Diagnostic Dashboard & Predictive Models